Program : **Diploma in Civil and Environmental Engineering**	
Course Code: **6352D**	Course Title: **Water Resources Engineering**
Semester: **6**	Credits: **4**
Course Category: **Open Elective**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To impart knowledge about the concepts of hydrologic cycle
- To facilitate the students with the basic understanding about water management.
- To summarize basic methods of irrigation and introduce the purposes of various hydraulic structures.

Course Prerequisites: Nil

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Familiarize with Surface and ground water resources and understand the concepts of Hydrologic cycle.	15	Understanding
CO2	Outline concepts of Water Management and Conservation system and drought and flood management strategies	14	Understanding
CO3	Understand basics of ground water and river engineering	13	Understanding
CO4	Outline the basics of Irrigation Engineering and the major and minor irrigation structures	16	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Familiarize with Surface and ground water resources and understand the concepts of Hydrologic cycle.**		
M1.01	Identify the different sources of water	1	Understanding
M1.02	Understand the rainfall distribution in India	1	Understanding
M1.03	Explain the terms like precipitation, evaporation and transpiration	1	Understanding
M1.04	Understand the concept of rainwater measurement using rain gauges	3	Understanding
M1.05	Understand the representation of rainfall data using mass curve and hyetograph.	3	Understanding
M1.06	Describe the basic concepts of Hydrograph analysis	3	Understanding
M1.07	Explain the concept of infiltration	2	Understanding
M1.08	Understand the factors affecting run off	1	Understanding

Contents

Surface and Ground Water Resources–Different sources of water - Rainfall distribution in India

Hydrologic cycle- Hydrologic cycle - Precipitation, Evaporation and Transpiration.Precipitation-mechanism, types, forms and measurement using rain gauges -

<table>
<tr><td colspan="4">Optimum number of rain gauges- representation of rainfall data-mass curve and hyetograph.
Runoff-components of runoff- Hydrograph analysis-Hydrograph from isolated storm-Base flow separation. Unit hydrograph – uses, assumptions and limitations of unit hydrograph theory.
Infiltration-measurement by double ring infiltrometer. Runoff - factors affecting runoff</td></tr>
<tr><td>CO2</td><td colspan="3">Outline concepts of Water Management and Conservation system and drought and flood management strategies</td></tr>
<tr><td>M2.01</td><td>Describe planning of water resources</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.02</td><td>Understand the need for conservation of water and management</td><td>1</td><td>Understanding</td></tr>
<tr><td>M2.03</td><td>Describe the role of forest in conserving water</td><td>1</td><td>Understanding</td></tr>
<tr><td>M2.04</td><td>Understand the rain water collection and storage system.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.05</td><td>Understand river water storage by check dams</td><td>2</td><td>Understanding</td></tr>
<tr><td>M2.06</td><td>Outline the drought and flood management strategies</td><td>4</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents
Description of water resources planning-National water policy. Planning arrangements for drinking water supply project - irrigation water supply project, hydropower generation project and flood control project
Water Management Conservation system: Water conservation and management- surface water, rain water and ground water. Methods for improving percolation of water – Role of forest in conserving water – Rain water harvesting systems – Roof water collection system – Surface runoff collection – Ground water recharge – Rain pits – Terracing – Contour bunding – Check Dams – Functions and Construction details.Water Quality management-fresh water management, recycling and reuse of water.
Drought and Flood Management: Drought and floods ~ Drought and flood affected regions of India ~ Definition of drought ~ Tackling drought through water management ~ Flood management measures.</td></tr>
<tr><td>CO3</td><td colspan="3">Understand basics of ground water and river engineering</td></tr>
<tr><td>M3.01</td><td>Understand the concept of ground water recharge</td><td>2</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Identify the types of aquifers and wells</td><td>3</td><td>Understanding</td></tr>
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M3.03	Understand the concept of computation of yield of an open well	3	Understanding
M3.04	Understand the concepts of meandering and river training	3	Understanding
M3.05	Outline the various types of reservoirs	2	Understanding
Contents: **Groundwater** –vertical distribution of groundwater-Types of aquifer-Aquifer properties-Darcy's law-Steady radial flow to a well-unconfined and confined aquifers-Types of wells-open well, artesian well and tube well-Estimation of yield of an open well (Concepts only)-Types of tube wells. **River Engineering**-meandering-rivertraining –objectives, classification, river training methods-levees, guide banks, groynes, artificial cut-offs, pitching, pitched islands. Reservoir-various types-zones of storages			
CO4	**Outline the basics of Irrigation Engineering and the major and minor irrigation structures**		
M4.01	Identify the types of irrigation systems.	2	Understanding
M4.02	Understand the terms crop period, base period, duty, delta, irrigation efficiencies and intensity of irrigation	2	Understanding
M4.03	Enumerate the methods of improving and factors affecting duty.	2	Understanding
M4.04	Classify the irrigation projects	2	Understanding
M4.05	Describe the classification of canals	2	Understanding
M4.06	Outline the various irrigation structures	3	Understanding
M4.07	Understand the need and classification of cross drainage works	3	Understanding
Contents: **Irrigation**– Necessity, Benefits and ill effects. Types: flow and lift irrigation - perennial and inundation irrigation. Soil-water –plant relationships, Irrigation efficiencies, Crop period, base period, Duty, Delta, Command area, intensity of irrigation, duty-factors affecting and method of improving duty. **Classification of irrigation projects**-Major, Medium and Minor schemes (Definition only) **Canals** - Classification according to alignment and position in the canal network.Canal			

lining - Purpose, material used and its properties, advantages. **Irrigation structures** - classification of head works - storage and diversion head works - their suitability under different conditions. **Storage Headworks**: Dams and its classification, **Diversion head works** - Layout, components and their function. **Cross Drainage works** - Aqueduct, siphon aqueduct, super passage, level crossing (Brief description only)

Text / Reference:

T/R	Book Title/Author
T1	B.C.Punmia& BB Pande, "Irrigation and Water Power Engineering'', LaxmiPublications (P) Ltd.
T2	S.K.Garg, "Hydrology & Water Resources Engineering.", Khanna Publications, Delhi.
R3	P.N.Modi, "Irrigation, Water Resources & Water Power Engineering"-,S.B.H Publishers and Distributors, New Delhi.
R4	K.Subramanya,''Engineering Hydrology'', Tata Mc Graw Hill Series, New Delhi..
R5	M.J.Deodhar, "Elementary Engineering Hydrology" – Pearson education -Delhi
R6	R.K. Sharma, "Hydrology & Water Resources", Dhanpat Rai & Sons, Delhi.
R7	J.V.S. Moorthy, "Watershed management in India", Willey Eastern Ltd.
R8	C. Satyanarayan Murthy, "Water Resources Engineering Principles & Practice", New Age International Ltd., New Delhi
R9	K.R. Karanth, "Ground water assessment, Development & management", Tata Mc Graw Hill Pub. Co. Ltd., New Delhi.

Online Resources:

Sl.No.	Website Link
1	https://archive.nptel.ac.in/courses/105/105/105105110/